

思考 3 P111

~~$0 < R < 2$ $\because z=3$ 时发散~~

$R=1$ 当 $z=\frac{1}{2}$, $r=\frac{1}{2}$ 收敛

当 $z=3$, $r=2$ 发散

思考 7 P112

~~$a_n = \frac{1}{n!} (\cos z)^{(n)}$~~ 奇点: $\frac{z}{2} + k\pi$

$\therefore R = \frac{\pi}{2}$

3(3) $\sum_{n=1}^{\infty} n z^n$ $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)z}{n} = z$
 $\therefore R=1$, $|z| < 1$

3(5) $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{z}{e} = 1 \Rightarrow R=e$, $|z| < e$

3(7) $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1) \lg(n+1)}{n \lg n} \frac{z}{(n+1)} = \lim_{n \rightarrow \infty} \frac{\lg(n+1)}{n \lg n} z = 0$

$\therefore R \rightarrow +\infty$

~~$z \in \mathbb{C}$~~

4(2) $e^z = \sum_{n=0}^{+\infty} \frac{z^n}{n!}$ ~~$R \rightarrow +\infty$~~ $R \rightarrow +\infty$

$e^z = e e^{z-1} = \sum_{n=0}^{+\infty} \frac{(z-1)^n}{n!}$, ~~$z \in \mathbb{C}$~~ $z \in \mathbb{C}$

$$4(4) \frac{1}{(z-2)^2} = \left(-\frac{1}{(z-2)} \right)' = \left(-\frac{1}{(z-1)-1} \right)' = \left(\sum_{n=0}^{\infty} (z-1)^n \right)' = \sum_{n=0}^{\infty} n(z-1)^{n-1} \quad |z-1| < 1$$

$$4(5) \text{ LHS} = \frac{1}{9} \frac{1}{1 + (\frac{z-1}{3})^2} = \frac{1}{9} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-1}{3} \right)^{2n}$$

$$\left(\frac{z-1}{3} \right)^2 < 1, z < z < 4$$

$$7. \left| \frac{C_{n+1}}{C_n} \right| = R \Rightarrow R = \left| \frac{C_n}{C_{n+1}} \right|$$

$$R' = \frac{\operatorname{Re}(C_n)}{\operatorname{Re}(C_{n+1})} = \frac{\operatorname{Re}(R C_{n+1})}{\operatorname{Re}(C_{n+1})}$$

当 $z < R$

$$\sum_{n=0}^{\infty} C_n z^n \text{ 收敛} \rightarrow \sum_{n=0}^{\infty} |C_n| |z|^n \text{ 收敛} \xrightarrow{|C_n| > \operatorname{Re}(C_n)} \sum_{n=0}^{\infty} \operatorname{Re}(C_n) |z|^n \text{ 收敛}$$

∴ 得证

$$11. f(z) = f(-z)$$

$$f(z) - f(-z) = \sum_{k=0}^{\infty} C_k (z^k - (-z)^k) = 0$$

当 $k = 2n+1$, $z^k - (-z)^k \neq 0$ 且全部同号

$$\therefore C_{2n+1} = 0$$

12.

12.

~~(1)~~ (1)

$$\text{RHS} = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{z^n e^{z\zeta}}{n! \zeta^{n+1}} d\zeta = \frac{1}{n!} \left(\frac{z^n e^{z\zeta}}{n!} \right)^{(n)} = \frac{z^n}{(n!)^2} \cdot (e^{z\zeta})^{(n)}$$

$$= \frac{z^n \cdot z^n}{(n!)^2} = \text{LHS} \text{ 得证}$$

(2) LHS = $\sum_{n=0}^{\infty} \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{z^n e^{z\zeta}}{n! \zeta^{n+1}} d\zeta$ 取 $\zeta = \cos\theta$, $\theta \in (0, 2\pi)$

~~$$\frac{1}{2\pi i} \sum_{n=0}^{\infty} \oint_{|\zeta|=1} \frac{z^n e^{z\cos\theta}}{n! (\cos\theta)^{n+1}} (-\sin\theta) d\theta = \frac{1}{2\pi i} \sum_{n=0}^{\infty} \frac{z^n e^{z\cos\theta}}{n! (\cos\theta)^{n+1}}$$~~

$$= \frac{1}{2\pi i} \sum_{n=0}^{\infty} \oint \frac{\left(\frac{z}{\zeta}\right)^n e^{z\zeta}}{n! \zeta} d\zeta \quad \text{令 } t = \frac{z}{\zeta}$$

~~$$\neq \frac{1}{2\pi i} \sum_{n=0}^{\infty} \oint \frac{t^n}{n!} e^{\frac{t\zeta}{\zeta}} \frac{d\zeta}{\zeta^2}$$~~

$$= \frac{1}{2\pi i} \oint \sum_{n=0}^{\infty} \frac{\left(\frac{z}{\zeta}\right)^n}{n!} \cdot \frac{e^{z\zeta}}{\zeta} d\zeta$$

$$= \frac{1}{2\pi i} \oint e^{\frac{z}{\zeta}} \cdot e^{z\zeta} \frac{d\zeta}{\zeta}$$

$$= \frac{1}{2\pi i} \oint e^{z(\zeta + \frac{1}{\zeta})} \frac{d\zeta}{\zeta} \quad \leftarrow \text{取 } \zeta = e^{i\theta}$$

$$= \frac{1}{2\pi} \oint e^{2z\cos\theta} d\theta$$